Since $f'(x) = 1/x$, we have, for $x = 1$,

$$1 = \lim_{h \to 0} \ln (1 + h)^{1/h}.$$

We next observe, from Theorem (7.19), that

$$(1 + h)^{1/h} = e^{\ln (1 + h)^{1/h}}.$$

Since the natural exponential function is continuous at 1, it follows from Theorem (2.25) that

$$\lim_{h \to 0} (1 + h)^{1/h} = \lim_{h \to 0} \left[e^{\ln (1 + h)^{1/h}}\right] = e^{\left[\lim_{h \to 0} \ln (1 + h)^{1/h}\right]} = e^1 = e.$$

This establishes part (i) of the theorem. The limit in part (ii) may be obtained by introducing the change of variable $n = 1/h$ with $h > 0$. ■

The formulas in Theorem (7.32) are sometimes used to *define* the number $e$. You may find it instructive to calculate $(1 + h)^{1/h}$ for numerically small values of $h$. Some approximate values are given in the following table.

| $h$ | $(1 + h)^{1/h}$ | $h$ | $(1 + h)^{1/h}$ |
|---|---|---|---|
| 0.01 | 2.704814 | −0.01 | 2.731999 |
| 0.001 | 2.716924 | −0.001 | 2.719642 |
| 0.0001 | 2.718146 | −0.0001 | 2.718418 |
| 0.00001 | 2.718268 | −0.00001 | 2.718295 |
| 0.000001 | 2.718280 | −0.000001 | 2.718283 |

To five decimal places, $e \approx 2.71828$.

## EXERCISES 7.5

**Exer. 1–26: Find $f'(x)$ if $f(x)$ is the given expression.**

1 $7^x$

2 $5^{-x}$

3 $8^{x^2+1}$

4 $9^{\sqrt{x}}$

5 $\log (x^4 + 3x^2 + 1)$

6 $\log_3 |6x - 7|$

7 $5^{3x-4}$

8 $3^{2-x^2}$

9 $(x^2 + 1)10^{1/x}$

10 $(10^x + 10^{-x})^{10}$

11 $\log (3x^2 + 2)^5$

12 $\log \sqrt{x^2 + 1}$

13 $\log_5 \left|\dfrac{6x + 4}{2x - 3}\right|$

14 $\log \left|\dfrac{1 - x^2}{2 - 5x^3}\right|$

15 $\log \ln x$

16 $\ln \log x$

17 $x^e + e^x$

18 $x^\pi \pi^x$

19 $(x + 1)^x$

20 $x^{4+x^2}$

21 $2^{\sin^2 x}$

22 $4^{\sec 3x}$

23 $x^{\tan x}$

24 $(\cos 2x)^x$

25 (a) $e^e$ (b) $x^5$ (c) $x^{\sqrt{5}}$ (d) $(\sqrt{5})^x$ (e) $x^{(x^2)}$

26 (a) $\pi^\pi$ (b) $x^4$ (c) $x^\pi$ (d) $\pi^x$ (e) $x^{2x}$

**Exer. 27–42: Evaluate the integral.**

27 (a) $\int 7^x \, dx$ (b) $\int_{-2}^{1} 7^x \, dx$

28 (a) $\int 3^x \, dx$ (b) $\int_{-1}^{0} 3^x \, dx$

29 (a) $\int 5^{-2x} \, dx$ (b) $\int_{1}^{2} 5^{-2x} \, dx$

30 (a) $\int 2^{3x-1} \, dx$ (b) $\int_{-1}^{1} 2^{3x-1} \, dx$

31 $\int 10^{3x} \, dx$

32 $\int 5^{-5x} \, dx$

33 $\int x(3^{-x^2}) \, dx$

34 $\int \dfrac{(2^x + 1)^2}{2^x} \, dx$

35 $\int \frac{2^x}{2^x + 1}\,dx$

36 $\int \frac{3^x}{\sqrt{3^x + 4}}\,dx$

37 $\int \frac{1}{x \log x}\,dx$

38 $\int \frac{10^{\sqrt{x}}}{\sqrt{x}}\,dx$

39 $\int 3^{\cos x} \sin x\,dx$

40 $\int \frac{5^{\tan x}}{\cos^2 x}\,dx$

41 (a) $\int \pi^\pi\,dx$ (b) $\int x^4\,dx$

(c) $\int x^\pi\,dx$ (d) $\int \pi^x\,dx$

42 (a) $\int e^e\,dx$ (b) $\int x^5\,dx$

(c) $\int x^{\sqrt{5}}\,dx$ (d) $\int (\sqrt{5})^x\,dx$

43 Find the area of the region bounded by the graphs of $y = 2^x$, $x + y = 1$, and $x = 1$.

44 The region under the graph of $y = 3^{-x}$ from $x = 1$ to $x = 2$ is revolved about the $x$-axis. Find the volume of the resulting solid.

45 An economist predicts that the buying power $B(t)$ of a dollar $t$ years from now will decrease according to the formula $B(t) = (0.95)^t$.

(a) At approximately what rate will the buying power be decreasing two years from now?

(b) Estimate the *average buying power* of the dollar over the next two years.

46 When a person takes a 100-milligram tablet of an asthma drug orally, the rate $R$ at which the drug enters the bloodstream is predicted to be $R = 5(0.95)^t$ mg/min. If the bloodstream does not contain any trace of the drug when the tablet is taken, determine the number of minutes needed for 50 milligrams to enter the bloodstream.

47 One thousand trout, each one year old, are introduced into a large pond. The number still alive after $t$ years is predicted to be $N(t) = 1000(0.9)^t$.

(a) Approximate the death rate $dN/dt$ at times $t = 1$ and $t = 5$. At what rate is the population decreasing when $N = 500$?

(b) The weight $W(t)$ (in pounds) of an individual trout is expected to increase according to the formula $W(t) = 0.2 + 1.5t$. After approximately how many years is the total number of pounds of trout in the pond a maximum?

48 The vapor pressure $P$ (in psi), a measure of the volatility of a liquid, is related to its temperature $T$ (in °F) by the *Antoine equation*: $\log P = a + [b/(c + T)]$, for constants $a$, $b$, and $c$. Vapor pressure increases rapidly with an increase in temperature. Find conditions on $a$, $b$, and $c$ that guarantee that $P$ is an increasing function of $T$.

49 Chemists use a number denoted by pH to describe quantitatively the acidity or basicity of solutions. By definition, $\text{pH} = -\log [\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen ion concentration in moles per liter. For a certain brand of vinegar it is estimated (with a maximum percentage error of $\pm 0.5\%$) that $[\text{H}^+] \approx 6.3 \times 10^{-3}$. Calculate the pH and use differentials to estimate the maximum percentage error in the calculation.

50 The magnitude $R$ (on the Richter scale) of an earthquake of intensity $I$ may be found by means of the formula $R = \log (I/I_0)$, where $I_0$ is a certain minimum intensity. Suppose the intensity of an earthquake is estimated to be 100 times $I_0$. If the maximum percentage error in the estimate is $\pm 1\%$, use differentials to approximate the maximum percentage error in the calculated value of $R$.

51 Let $R(x)$ be the reaction of a subject to a stimulus of strength $x$. For example, if the stimulus $x$ is *saltiness* (in grams of salt/liter), $R(x)$ may be the subject's estimate of how salty the solution tasted on a scale from 0 to 10. A function that has been proposed to relate $R$ to $x$ is given by the *Weber-Fechner formula*: $R = a \log (x/x_0)$, where $a$ is a positive constant.

(a) Show that $R = 0$ for the *threshold stimulus* $x = x_0$.

(b) The derivative $S = dR/dx$ is the *sensitivity* at stimulus level $x$ and measures the ability to detect small changes in stimulus level. Show that $S$ is inversely proportional to $x$, and compare $S(x)$ to $S(2x)$.

52 The loudness of sound, as experienced by the human ear, is based on intensity level. A formula used for finding the intensity level $\alpha$ that corresponds to a sound intensity $I$ is $\alpha = 10 \log (I/I_0)$ decibels, where $I_0$ is a special value of $I$ agreed to be the weakest sound that can be detected by the ear under certain conditions. Find the rate of change of $\alpha$ with respect to $I$ if

(a) $I$ is 10 times as great as $I_0$

(b) $I$ is 1000 times as great as $I_0$

(c) $I$ is 10,000 times as great as $I_0$ (This is the intensity level of the average voice.)

53 If a principal of $P$ dollars is invested in a savings account for $t$ years and the yearly interest rate $r$ (expressed as a decimal) is compounded $n$ times per year, then the amount $A$ in the account after $t$ years is given by the *compound interest formula*: $A = P[1 + (r/n)]^{nt}$.

(a) Let $h = r/n$ and show that

$$\ln A = \ln P + rt \ln (1 + h)^{1/h}.$$

(b) Let $n \to \infty$ and use the expression in part (a) to establish the formula $A = Pe^{rt}$ for interest *compounded continuously*.

54 Establish Theorem (7.32)(ii) by using the limit in part (i) and the change of variable $n = 1/h$.